

Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation and formatting followed throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into Dataset, Training, Flask, Templates and Model folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as Country, Expected_Years_Schooling, GNI_Per_Capita clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like home(), prediction(), predict() are meaningful and descriptive.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments explaining functionality.
6	Modular Design	Code is split into reusable functions/modules	Yes	Model training, Flask application, dataset and HTML templates are separated into different modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code removed during development.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation implemented. Flask handles invalid requests; advanced exception handling can be added.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Linear Regression Model	Python (Scikit-Learn)	Prediction Module	High
2	Flask Backend	Python, Flask	Web Application	High
3	HTML Templates	HTML, CSS	User Interface Pages	High
4	Label Encoder	Scikit-Learn	Country Encoding	Medium
5	HDI Dataset	CSV	Model Training & Prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with separate folders for datasets, training notebook, model, and Flask application.
Readability	5	Meaningful variable names, proper indentation and comments improve readability.
Reusability	5	Modular components such as trained model, Flask routes and HTML templates can be reused in future projects.
Documentation / Comments	5	Adequate comments provided for important code sections. Additional documentation can further improve maintainability.
Overall Score	5/5	The project follows good coding practices with clean structure, readability and reusable components. The project follows good coding practices with clean structure, readability and reusable components.